# AGENTIC MULTIMODAL RETRIEVAL-AUGMENTED GENERATION SYSTEM

CLIP · FAISS · LANGCHAIN · VLM · Neo4j

**Deep Neural Networks (AI-341)**
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# 1 Abstract

This proposal presents the design and implementation of an **Agentic Multimodal Retrieval-Augmented Generation (RAG)** system — a cutting-edge architecture that fuses visual understanding, semantic retrieval, structured knowledge graph reasoning, and autonomous agent orchestration into a single coherent pipeline. The system enables users to query a heterogeneous database of images, documents, and relational knowledge using natural language. An intelligent LangChain ReAct agent decides — in real time — whether to perform a visual search, a textual search, a knowledge graph traversal, or a hybrid combination, before synthesizing a grounded, evidence-backed answer via a Vision-Language Model (VLM).

The system introduces four key technical contributions beyond a standard RAG pipeline: *semantic chunking* for context-preserving text segmentation, *HNSW-indexed* vector retrieval for scalable approximate nearest-neighbor search, a *Neo4j knowledge graph* for structured factual reasoning, and *Reciprocal Rank Fusion (RRF)* for principled multi-source result merging. Together, these components place this project at the frontier of applied deep learning, multimodal AI, and agentic systems research.

# 2 Problem Statement & Motivation

Modern knowledge repositories — enterprise databases, industrial inspection archives, scientific literature collections — store data in heterogeneous, mixed-media formats: PDFs with embedded figures, image galleries with associated captions, and annotated documents. Conventional retrieval systems fail to bridge the semantic gap between modalities, creating four critical limitations:

- **Unimodal retrieval:** Systems cannot match a natural language query to a visually relevant image, nor retrieve textual passages based on visual content.
- **Static pipelines:** Fixed retrieval logic is applied regardless of query intent — a system designed for image search will perform poorly on predominantly textual queries.
- **No reasoning layer:** Raw retrieval results are returned without synthesis. The user receives a ranked list, not an answer.
- **No structured factual grounding:** Vector similarity cannot capture entity relationships, categorical hierarchies, or causal links — knowledge that is best represented as a graph.

**Motivating Example**

A quality engineer searches a manufacturing defect archive with the query: *"Show me images of surface cracks near welding joints and summarize the inspection notes, including which machine operators were responsible."* A unimodal text system returns notes but no images. A vector-only system cannot resolve the relational query about operators. **Our proposed system** routes this query through hybrid retrieval: HNSW-indexed CLIP search returns relevant images, semantic chunks retrieve inspection notes, and the Neo4j knowledge graph resolves the entity relationship between defect records and operators — all fused via RRF before the VLM synthesizes a final answer.

# 3 Project Objectives

1. Build a dual-modal embedding pipeline using OpenAI CLIP (ViT-B/32) for joint image–text encoding.

2. Index the multimodal knowledge base using FAISS with an **HNSW index** for scalable, high-speed approximate nearest-neighbor retrieval.
3. Implement **semantic chunking** to split documents into semantically coherent units rather than fixed-size token windows.
4. Construct a **Neo4j knowledge graph** storing entity relationships, metadata, and factual associations to support structured reasoning queries.
5. Implement a LangChain ReAct agent equipped with four custom tools: `VisualSearch`, `TextSearch`, `GraphSearch`, and `HybridSearch`.
6. Apply **Reciprocal Rank Fusion (RRF)** to merge ranked result lists from multiple retrieval sources into a single unified ranking.
7. Integrate a VLM (BLIP-2 or LLaVA) to perform deep visual reasoning on retrieved image evidence.
8. Develop a Gradio-based demo UI and evaluate using Recall@K, MRR, RRF gain, and qualitative reasoning quality metrics.

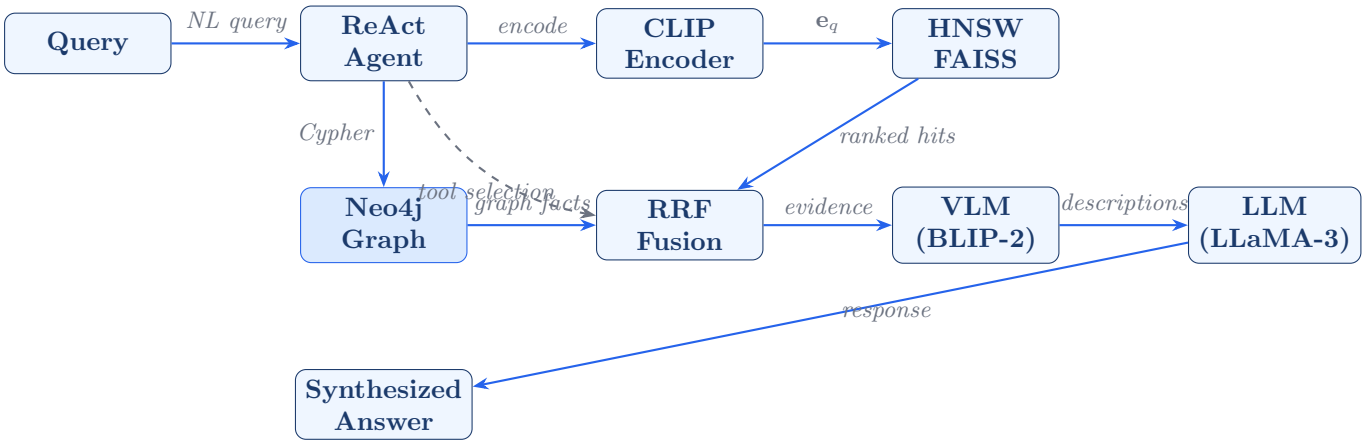
# 4 Proposed System Workflow

The system operates as a nine-stage agentic pipeline:

1. User submits a natural language query $q$ via the Gradio interface.
2. Query is encoded into a joint embedding $\mathbf{e}_q = g(q) \in \mathbb{R}^{512}$ using CLIP's text encoder.
3. The LangChain ReAct agent analyzes query intent and selects a retrieval strategy (visual, textual, graph, or hybrid).
4. For visual/textual queries: relevant items are retrieved from the HNSW-indexed FAISS store via approximate nearest-neighbor search.
5. For factual/relational queries: the agent issues a Cypher query to the Neo4j knowledge graph and retrieves structured entity data.
6. Results from multiple retrievers are merged using **Reciprocal Rank Fusion** into a single ranked evidence list.
7. If images are present in the evidence, the VLM processes each to produce a natural-language description.
8. All evidence (retrieved text chunks + VLM descriptions + graph facts) is passed to the LLM for synthesis.
9. A final grounded, evidence-backed response is returned to the user.

# 5 System Architecture

## 5.1 Architecture Diagram




## 5.2 Sub-Module Descriptions

### Semantic Chunking Module

Raw documents are split into semantically coherent chunks using embedding-based similarity rather than fixed token windows. Consecutive sentences are grouped as long as their cosine similarity exceeds a threshold $\delta$; a new chunk begins when semantic drift is detected. This preserves contextual integrity across chunk boundaries.

### Multimodal Embedding Module

CLIP encodes both images and semantically chunked text into a shared 512-dimensional embedding space using contrastive pretraining on 400M image–caption pairs. All database assets are encoded offline and stored persistently in the HNSW index.

### HNSW Vector Index (FAISS)

The Hierarchical Navigable Small World (HNSW) graph index replaces flat IVF for the primary vector store. HNSW builds a multi-layer proximity graph enabling $O(\log n)$ query complexity with superior recall–speed trade-offs compared to IVF-Flat at scale.

### Neo4j Knowledge Graph Module

A property graph database stores entities (images, documents, persons, objects, events) and their typed relationships. The ReAct agent issues Cypher queries for structured lookups — e.g., finding all defect records associated with a specific machine or operator — which vector similarity alone cannot resolve.

### RRF Fusion Module

Results from the HNSW vector retriever, the textual retriever, and the Neo4j graph retriever are independently ranked and then fused via Reciprocal Rank Fusion into a single unified evidence list before being passed to the VLM and LLM.

### Agentic Orchestration Layer

A LangChain ReAct agent reasons across multiple thought–action–observation cycles, dynamically selecting among four tools (`VisualSearch`, `TextSearch`, `GraphSearch`, `HybridSearch`) and iterating until sufficient evidence is gathered.

### Vision-Language Reasoning Module

BLIP-2 or LLaVA performs visual question answering on retrieved images, converting raw

visual evidence into textual descriptions that the agent can incorporate into its final synthesis.

**Gradio Frontend**

A clean interface displays the agent's tool selection trace, retrieved evidence (images + text + graph facts) with scores, and the final synthesized answer.

# 6 Technical Components & Mathematical Formulation

## 6.1 CLIP Cross-Modal Embedding

CLIP encodes images $x_i$ and text $x_t$ into a shared embedding space via encoders $f(\cdot)$ and $g(\cdot)$:

$$\mathrm{sim}(x_i, x_t) = \frac{f(x_i) \cdot g(x_t)}{\|f(x_i)\| \, \|g(x_t)\|}$$

The model is trained with a symmetric cross-entropy loss over a batch of $N$ image–text pairs:

$$\mathcal{L}_{\mathrm{CLIP}} = -\frac{1}{2N} \sum_{k=1}^{N} \left[ \log \frac{e^{\mathrm{sim}(x_i^k, x_t^k)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(x_i^k, x_t^j)/\tau}} + \log \frac{e^{\mathrm{sim}(x_t^k, x_i^k)/\tau}}{\sum_{j=1}^{N} e^{\mathrm{sim}(x_t^k, x_i^j)/\tau}} \right]$$

where $\tau$ is a learnable temperature parameter.

## 6.2 Semantic Chunking

Unlike fixed-size token splitting, semantic chunking groups consecutive sentences $s_1, s_2, \ldots, s_n$ based on embedding similarity. A chunk boundary is inserted between $s_i$ and $s_{i+1}$ when:

$$\mathrm{sim}\big(g(s_i),\ g(s_{i+1})\big) < \delta$$

where $g(\cdot)$ is the sentence encoder and $\delta \in [0, 1]$ is a tunable semantic drift threshold. This ensures each chunk represents a coherent semantic unit, improving retrieval precision over naive token-window chunking.

## 6.3 HNSW Approximate Nearest-Neighbor Retrieval

The Hierarchical Navigable Small World (HNSW) index organizes vectors in a multi-layer proximity graph. Each layer $\ell$ contains a subset of nodes with long-range connections, enabling greedy graph traversal from coarse to fine resolution. Given query $\mathbf{q} \in \mathbb{R}^d$, retrieval proceeds as:

$$\mathcal{R}_K^{\mathrm{HNSW}}(\mathbf{q}) = \underset{\substack{K \\ x \in \mathcal{D}}}{\mathrm{argmin}} \, \|\mathbf{q} - \mathbf{e}_x\|_2$$

with query complexity $O(\log |\mathcal{D}|)$, compared to $O(\sqrt{|\mathcal{D}|})$ for IVF. HNSW is configured with parameters $M$ (max connections per node) and $ef_{\mathrm{construction}}$ (beam width during index build), providing a tunable recall–latency trade-off.

## 6.4 Neo4j Knowledge Graph Reasoning

The knowledge graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ stores entities $v \in \mathcal{V}$ (images, documents, persons, objects) and typed relationships $e \in \mathcal{E}$. For a structured query $q_s$ (e.g., *"which operator inspected defect D-42?"*), the agent issues a Cypher query:

```
MATCH (d:Defect {id: "D-42"})<-[:INSPECTED]-(op:Operator)
RETURN op.name, op.shift, d.severity
```

The graph path score between query entity $v_q$ and result entity $v_r$ is defined as:

$$s_{\text{graph}}(v_q, v_r) = \frac{1}{1 + d_{\mathcal{G}}(v_q, v_r)}$$

where $d_{\mathcal{G}}(\cdot, \cdot)$ is the shortest-path distance in $\mathcal{G}$. This score is passed to the RRF fusion module alongside vector similarity scores.

### 6.5 Reciprocal Rank Fusion

RRF merges ranked result lists $\{R_1, R_2, \ldots, R_m\}$ from $m$ independent retrievers (HNSW visual, HNSW textual, Neo4j graph) into a single unified ranking. For each document $d$, the fused score is:

$$\text{RRF}(d) = \sum_{i=1}^{m} \frac{1}{k + r_i(d)}$$

where $r_i(d)$ is the rank of document $d$ in retriever $R_i$, and $k$ is a smoothing constant (typically $k = 60$). Documents not returned by a retriever are assigned $r_i(d) = \infty$, contributing zero to the sum. RRF is parameter-light, robust to score scale differences across retrievers, and consistently outperforms linear score combination in multi-source retrieval settings [7].

### 6.6 Hybrid Score Fusion (Legacy)

For ablation comparison, a weighted linear combination of visual and textual scores is also maintained:

$$s_{\text{hybrid}}(x, q) = \alpha \cdot s_{\text{visual}}(x, q) + (1 - \alpha) \cdot s_{\text{text}}(x, q), \quad \alpha \in [0, 1]$$

RRF replaces this as the primary fusion method; the weighted fusion is retained for ablation study purposes.

### 6.7 ReAct Agent Reasoning

The agent follows an interleaved Thought–Action–Observation loop [4]. At each step $t$:

$$a_t = \pi_\theta(q, h_{<t}), \quad o_t = \text{Tool}(a_t), \quad h_t = (q, a_1, o_1, \ldots, a_t, o_t)$$

where $\pi_\theta$ is the LLM policy, $h_{<t}$ is the full interaction history, and $\text{Tool}(\cdot) \in \{\texttt{VisualSearch}, \texttt{TextSearch}, \texttt{Gra}$

### 6.8 VLM Visual Reasoning

For each retrieved image $x_i^*$, the VLM generates a textual description conditioned on both the image and the original query:

$$\hat{y}_i = \text{VLM}(x_i^*, q) = \arg\max_y P_\phi(y \mid x_i^*, q)$$

These descriptions $\{\hat{y}_i\}$, concatenated with retrieved text chunks and graph facts, are passed to the LLM for final answer synthesis.

# 7 Technology Stack

| Component | Technology | Justification |
|---|---|---|
| **Embedding Model** | CLIP ViT-B/32 (HuggingFace) | State-of-the-art cross-modal encoder; no GPU needed for inference |
| **Text Segmentation** | Semantic Chunking (SBERT) | Context-preserving splits via embedding similarity; outperforms fixed-window chunking |
| **Vector Index** | FAISS HNSW | $O(\log n)$ query complexity; superior recall–speed trade-off over IVF-Flat |
| **Metadata Store** | ChromaDB | Persistent metadata and chunk storage alongside FAISS |
| **Knowledge Graph** | Neo4j (Cypher) | Property graph DB for structured entity-relationship reasoning |
| **Result Fusion** | Reciprocal Rank Fusion (RRF) | Parameter-light, scale-invariant multi-retriever result merging |
| **Agent Framework** | LangChain ReAct | Production-grade orchestration with 4-tool agent and memory support |
| **VLM Reasoning** | BLIP-2 / LLaVA (Ollama) | Open-source VQA; runs on Colab T4 GPU within free-tier constraints |
| **LLM Backend** | Groq API (LLaMA-3-70B) | Free-tier API; extremely fast inference; strong instruction following |
| **Deep Learning** | PyTorch + HuggingFace | Standard DL stack; broad model ecosystem |
| **Frontend / Demo** | Gradio | Rapid UI prototyping; native HuggingFace Spaces deployment |
| **Compute** | Google Colab (T4 GPU) | Free GPU; sufficient for BLIP-2 inference and HNSW indexing |

# 8 Dataset

The system will be demonstrated on three complementary data sources, each targeting a different retrieval modality.

| Dataset | Size | Purpose |
|---|---|---|
| MS-COCO val2017 | 5,000 images / 25,000 captions | Primary visual + textual retrieval benchmark |
| MVTec AD | 5,354 images (15 categories) | Domain demo: industrial defect retrieval |
| ArXiv Abstracts | 1,000 semantic chunks | Textual retrieval; semantic chunking evaluation |
| Neo4j Graph | 10K nodes / 30K edges | Entity–relationship store for factual queries |

The Neo4j graph is constructed by extracting named entities and relationships from COCO captions and MVTec inspection metadata using an NLP pipeline (spaCy + custom relation extraction), then populating a property graph with typed edges (`CONTAINS`, `NEAR`, `INSPECTED_BY`, etc.).

# 9 Implementation Plan

| Phase | Task | Deliverable |
|---|---|---|
| **1** | Literature review & environment setup | All libraries installed; CLIP baseline running on Colab |
| **2** | Semantic chunking pipeline | Chunker producing coherent segments; threshold $\delta$ tuned |
| **3** | CLIP embedding + HNSW index | HNSW-indexed FAISS store; Recall@10 baseline measured |
| **4** | Neo4j graph construction | Entity graph populated; Cypher queries returning correct results |
| **5** | LangChain ReAct agent (4 tools) | Agent reasoning trace logged; all tools callable |
| **6** | RRF fusion module | Multi-retriever results merged; RRF gain over single-retriever measured |
| **7** | VLM integration | BLIP-2 / LLaVA answering image queries end-to-end |
| **8** | Gradio demo UI + evaluation | Complete demo live; MRR / Recall / RRF metrics reported |

# 10 Evaluation Plan

## 10.1 Quantitative Metrics

**Recall@K** Fraction of queries where the correct item appears in the top-$K$ retrieved results:

$$\text{Recall@K} = \frac{1}{|Q|} \sum_{q \in Q} \mathbf{1}[\text{relevant item} \in \mathcal{R}_K(q)]$$

Reported for $K \in \{1, 5, 10\}$ across visual, textual, and graph query types.

**Mean Reciprocal Rank (MRR)** Average reciprocal rank of the first relevant result:

$$\text{MRR} = \frac{1}{|Q|} \sum_{q \in Q} \frac{1}{\text{rank}_q}$$

**RRF Gain** Improvement in MRR when using RRF fusion versus the single best individual retriever:

$$\Delta_{\text{RRF}} = \text{MRR}_{\text{RRF}} - \max_i \text{MRR}_{R_i}$$

A positive $\Delta_{\text{RRF}}$ confirms that multi-source fusion adds measurable value.

**Chunking Quality (Coherence Score)** Average intra-chunk cosine similarity versus inter-chunk cosine similarity, measuring how well semantic boundaries are preserved:

$$C_{\text{chunk}} = \overline{\text{sim}_{\text{intra}}} - \overline{\text{sim}_{\text{inter}}}$$

Higher $C_{\text{chunk}}$ indicates cleaner semantic segmentation.

**Graph Query Accuracy** Proportion of structured (relational) queries where the Neo4j Cypher response contains the correct entity, evaluated against a manually annotated ground-truth set.

**VLM Answer Quality** BERTScore $F_1$ between generated image descriptions and reference COCO captions, measuring semantic fidelity of visual reasoning.

### 10.2 Ablation Study

To isolate the contribution of each new component, we run four ablation conditions:

1. **Base**: CLIP + IVF-Flat FAISS + fixed chunking (no Neo4j, no RRF)
2. **+HNSW**: Replace IVF-Flat with HNSW index
3. **+Semantic Chunking**: Add semantic chunking over fixed-window splitting
4. **+Neo4j + RRF**: Full system with knowledge graph and RRF fusion

MRR and Recall@10 are reported for each condition, demonstrating incremental improvement.

### 10.3 Qualitative Evaluation

A set of 50 manually curated test queries (15 visual, 15 textual, 10 hybrid, 10 graph/relational) will be evaluated by both authors using a 5-point Likert scale assessing answer relevance, grounding accuracy, and fluency.

## 11 Novelty & Research Contribution

**Key Contributions**

1. **Dynamic agentic retrieval routing:** Unlike static multimodal RAG pipelines, our agent autonomously selects among four retrieval modalities per query — including structured graph traversal, a capability absent in prior multimodal RAG work.
2. **Semantic chunking for multimodal RAG:** Replacing fixed token windows with

embedding-similarity-based segmentation preserves contextual coherence across chunk boundaries, improving both retrieval precision and VLM reasoning quality.

3. **HNSW for scalable multimodal retrieval:** Adopting HNSW over IVF-Flat delivers $O(\log n)$ query complexity and superior recall at scale, validated through ablation against the IVF baseline.
4. **Neo4j knowledge graph integration:** The system extends multimodal RAG with a structured factual reasoning layer, enabling entity-relationship queries that pure vector similarity cannot answer.
5. **RRF for principled multi-source fusion:** Reciprocal Rank Fusion replaces ad-hoc weighted score combination with a theoretically grounded, scale-invariant merging strategy, with measurable MRR gain quantified via $\Delta_{\text{RRF}}$.

## 12 Expected Outcomes

- A fully functional, demo-ready multimodal RAG system with Gradio interface supporting visual, textual, graph, and hybrid queries.
- GitHub repository with clean, documented, MIT-licensed codebase including HNSW indexer, semantic chunker, Neo4j graph builder, and RRF fusion module.
- Recall@10 $> 75\%$ on MS-COCO val2017 for both visual and textual query types.
- Positive RRF gain ($\Delta_{\text{RRF}} > 0$) over the single best retriever baseline.
- Graph query accuracy $> 80\%$ on the structured entity-relationship evaluation set.
- A comprehensive technical report documenting architecture, ablation studies, and findings.

## 13 Conclusion

This proposal outlines an Agentic Multimodal RAG system that substantially advances beyond conventional static retrieval pipelines. The four additions — semantic chunking, HNSW indexing, Neo4j knowledge graph integration, and Reciprocal Rank Fusion — each address a concrete limitation of baseline RAG: context fragmentation, retrieval latency, factual grounding gaps, and score-fusion brittleness respectively. Together, they produce a system whose design is grounded in recent seminal literature, feasible within the semester timeline on free compute, and of sufficient depth to serve as a strong research portfolio artifact.

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